

Scheme**of Studies**
BS in Mathematics

Category	Course Code	Course Title	Credit Hours	Pre-Requisite
GE-1	URCG-5118	Functional English	3(3-0)	Nil
GE-2	URCG-5105 URCG-5126	Islamic Studies (OR) Religious Education/Ethics	2(2-0)	Nil
GE-3	URCG-5123	Applications of Information and Communication Technologies (ICT)	3(2-1)	Nil
Major-1	MATH-5101	Calculus-I	3(3-0)	Nil
Major-2	MATH-5102	Set Theory and Mathematical Logic	3(3-0)	Nil
Major-3	MATH-5103	Vector and Tensor Analysis	3(3-0)	Nil
GE-4	URCG-5111	Translation of Holy Quran-I*	Non-Cr. Hour	Nil

Semester Total Credit Hours: 17

Semester-II

Category	Course Code	Course Title	Credit Hours	Pre-Requisite
GE-5	URCG-5112	Fables, Wisdom and EPICS	2(2-0)	Nil
GE-6	URCG-5116	Science of Society-I	2(2-0)	Nil
GE-7	URCG-5120	Exploring Quantitative Skills	3(3-0)	Nil
GE-8	URCG-5127	Seerat of the Holy Prophet (SAW)*	1(1-0)	Nil
Major-4	MATH-5104	Calculus-II	3(3-0)	MATH-5101
Major-5	MATH-5105	Linear Algebra	3(3-0)	Nil
Major-6	MATH-5106	Mechanics	3(3-0)	Nil

Semester Total Credit Hours: 17

Semester-III

Category	Course Code	Course Title	Credit Hours	Pre-Requisite
GE-9	URCG-5119	Expository Writing	3(3-0)	Nil
GE-10	URCG-5121	Tools for Quantitative Reasoning	3(3-0)	Nil
GE-11	URCG-5122	Ideology and Constitution of Pakistan	2(2-0)	Nil
Major-7	MATH-5107	Calculus-III	3(3-0)	MATH-5104
Major-8	MATH-5108	Algebra-I	3(3-0)	Nil
Major-9	MATH-5109	Ordinary Differential Equations	3(3-0)	Nil
GE-4	URCG-5111	Translation of Holy Quran-II*	Non-Cr. Hour	Nil

Semester Total Credit Hours: 17

Semester-IV

Category	Course Code	Course Title	Credit Hours	Pre-Requisite
GE-12	URCG-5114	Basic Science	3(2-1)	Nil
GE-13	URCG-5124	Entrepreneurship	2(2-0)	Nil
GE-14	URCG-5125	Civics and Community Engagement	2(2-0)	Nil
Major-10	MATH-5110	Algebra-II	3(3-0)	MATH-5108
Major-11	MATH-5111	Discrete Mathematics	3(3-0)	Nil
Major-12	MATH-5112	Number Theory	3(3-0)	Nil

Semester Total Credit Hours: 16

Total Credit Hours: 67

Semester-V

Category	Course Code	Course Title	Credit Hours	Pre-Requisite
Major-13	MATH-6101	Real Analysis-I	3(3-0)	Nil
Major-14	MATH-6102	Topology	3(3-0)	Nil
Major-15	MATH-6103	Differential Geometry	3(3-0)	Nil
Major-16	MATH-6104	Mathematical Methods	3(3-0)	Nil
Indn-01	STAT-6101	Introductory Statistics	3(3-0)	Nil
GE-4	URCG-5111	Translation of Holy Quran-III*	Non-Cr. Hour	Nil

Semester Total Credit Hours: 15

Semester-VI

Category	Course Code	Course Title	Credit Hours	Pre-Requisite
Major-17	MATH-6105	Real Analysis-II	3(3-0)	MATH-6101
Major-18	MATH-6106	Classical Mechanics	3(3-0)	Nil
Major-19	MATH-6107	Complex Analysis	3(3-0)	Nil
Major-20	MATH-6108	Functional Analysis	3(3-0)	Nil
Indn-02	STAT-6102	Introduction to Probability Distributions	3(3-0)	Nil

Semester Total Credit Hours: 15

Semester-VII

Category	Course Code	Course Title	Credit Hours	Pre-Requisite
Major-21	MATH-6109	Numerical Analysis-I	3(3-0)	Nil
Major-22	MATH-6110	Partial Differential Equations	3(3-0)	Nil
Major-23	MATH-61xx	Elective-I**	3(3-0)	Nil
Major-24	MATH-61xx	Elective-II**	3(3-0)	Nil
Indn-03	CSIT-6101	Programming Languages for Mathematicians	3(3-0)	Nil
GE-4	URCG-5111	Translation of Holy Quran-IV*	Non-Cr. Hour	Nil
Compulsory	MATH-6111	Field experience/internship	3(3-0)	Nil

Semester Total Credit Hours: 18

Semester-VIII

Category	Course Code	Course Title	Credit Hours	Pre-Requisite
Major-25	MATH-6112	Numerical Analysis-II	3(3-0)	MATH-6109
Major-26	MATH-6113	Integral Equations	3(3-0)	Nil
Major-27	MATH-61xx	Elective-III**	3(3-0)	Nil
Major-28	MATH-61xx	Elective-IV**	3(3-0)	Nil
Indn-04	CSEC-6102	Web System and Technology	3(3-0)	Nil
Compulsory	MATH-6114	Capstone project	3(3-0)	Nil

Semester Total Credit Hours: 18

Degree Program Total: 133

*These courses for Muslim students only.

** These four courses are optional & can be selected from the following list:

Note: These courses will be offered by the department from the list of concentration elective courses as per availability of the resources.